Estimated Time of Arrival (ETA) Performance System Comparative Evaluation

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MITRE

MITRE Work in the Field of Flight Management Systems and Computers

- ***2004 - Assessment of Operational Differences Among Flight Management Systems***
- ***2005 - Analysis of Advanced Flight Management Systems (FMSs)***
- ***2006 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Lateral Path***
- ***2007 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Vertical Path***
- ***2008 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Radius-to-Fix Path Terminators***
- ***2009 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Lateral and Vertical Path Integration***
- ***2010 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Standard Instrument Departures***
- ***2011 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Area Navigation (RNAV) Holding***
- ***2012 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Standard Instrument Departure with Radius-to-Fix Legs***
- ***2013 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Optimized Profile Descents***
- ***2014 - Analysis of Flight Management Systems (FMSs), FMC Field Observations Trial, Performance Based Navigation to xLS Landing System (PBN to xLS)***

Note: All these reports were presented as papers to the American Institute of Aeronautics and Astronautics (AIAA) / Institute of Electrical and Electronics Engineers (IEEE), annual Digital Avionics Systems Conferences (DASC). 5 papers have been awarded "Best Paper in Session" including back-to-back, 2012 and 2013.

The Performance Evaluation and Comparison

- **In 2012 RTCA formed SC-227 to update the *Minimum Aviation System Performance Standards (MASPS): Required Navigation Performance (RNP) for Area Navigation (RNAV), DO-236B*, based on lessons learned from implementing PBN. DO-236C completed 2013**
- **SC-227 was also asked for a forward looking update of the time-based requirements. In 2014 RNP RNAV system performance standard Change 1 to DO-236C added minimum requirements for system computed Estimated Time of Arrival (ETA).**
- **The new minimum <u>functional</u> requirement is that the ETA must be available for <u>each</u> fix in a flight plan (not just the “go to” fix).**
- **Importantly, Change 1 sets a minimum <u>performance</u> requirement on ETAs at all planned (future) fixes in the onboard flight profile.**
- **Thus the standard has been expanded both in the functional and the performance requirements for ETA in an RNP RNAV system.**

ETA Requirements from DO236C, Change 1

- Maximum Time Estimation Error (TEE) shall be less than 1% of time remaining to the fix or 10 seconds, whichever is greater, in the absence of error in forecast environmental conditions;
- Computation of ETAs for all downstream waypoints shall be complete within 30 seconds of entry of the appropriate data

Trial Plan Summary to the Manufacturers

- **The test profile was developed to fit the definition of the "reasonable" flight profile noted in DO-236C Change 1 requirements (25 fixes, approximately 500 NM).**
- **A flight plan from Salt Lake City to Denver was selected. It includes a complex SID, the KSLC.RWY35.NSIGN3.EKR; an approximately 1 hour cruise phase with less than 5 fixes, and a reasonably constrained (speeds and altitudes) STAR, (the KDEN.FRNCH3; and approach, the RNAV (RNP) Z RWY 35L.**
- **These procedures contain examples of both climbing and descending turns which will test the ability of the systems to appropriately predict ETAs using a fully integrated 3-D profile.**
- **There will also be an environmental wind (constant 300°/35 knots) that will be known precisely to the system at whatever points it can model (no uncertainty).**

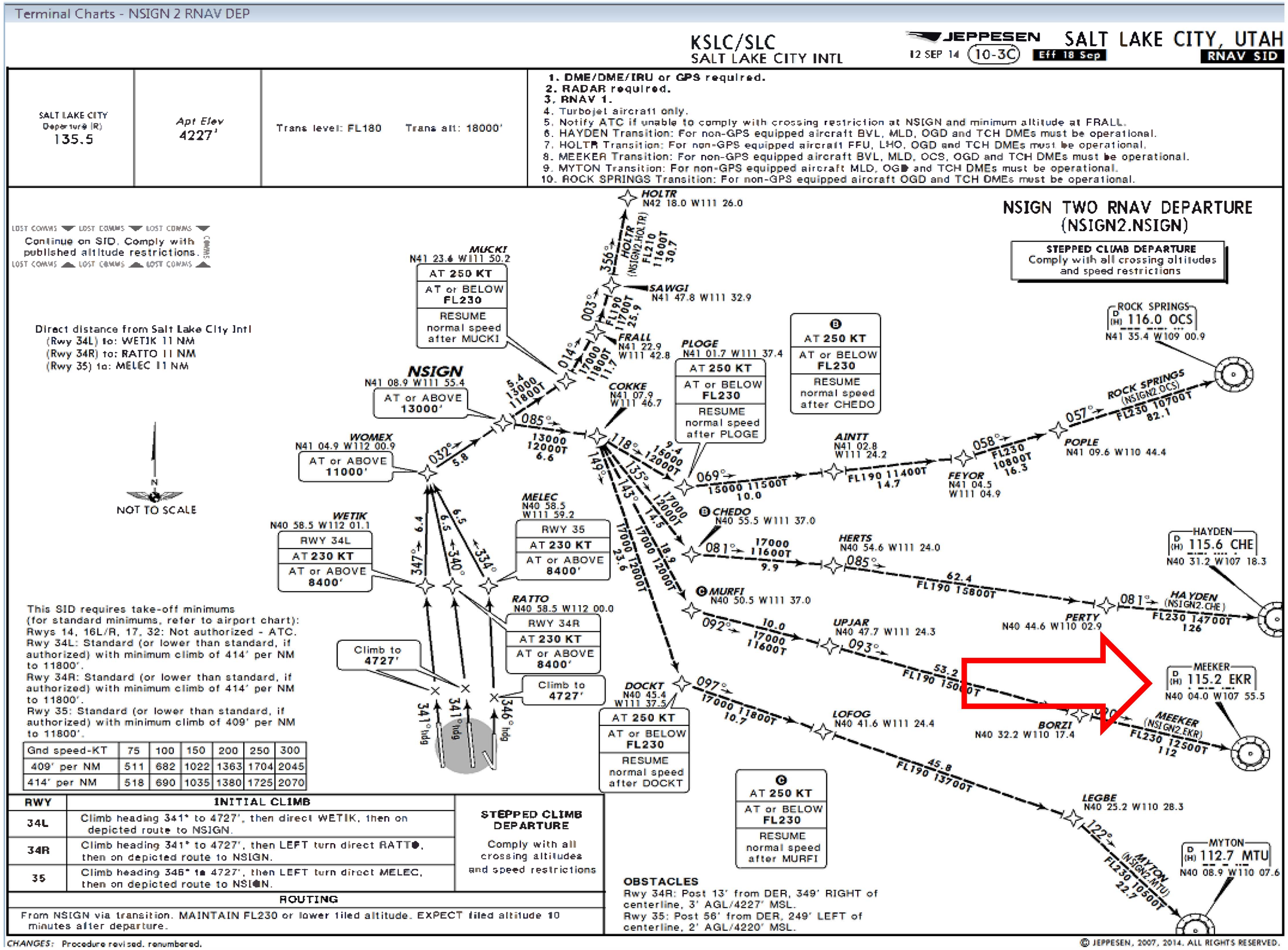
2014 Manufacturers and FMCs

Manufacturer	FMC	Aircraft
Honeywell (Phoenix)	787-8 BP1	B787 System Integration Test Station (SITS)
GE Aviation	U11.0	B737-700 Test Bench
Thalès (Toulouse)	FMS2 R1-A	Airbus 320 Test Bench
Rockwell Collins	FMS-6000	CL-604 Test Bench
Universal Avionics	UNS-1Ew SCN 1000.6	Cessna Citation II Test Bench
CMC Electronics	CMA-9000	Airbus 310 Test Bench
Garmin	G5000	Cessna Citation X+ and Sovereign+ Test Benches

Statement to Manufacturers

No FMS manufacturer or airline data will be identified in any report generated as a result of these evaluations, and draft reports will be sent to applicable FMS manufacturers and airlines for approval and edits or suggestions prior to release as a deliverable to the FAA or as a paper presented at DASC.

2015 – KSLC NSIGN TWO RNAV DEPARTURE



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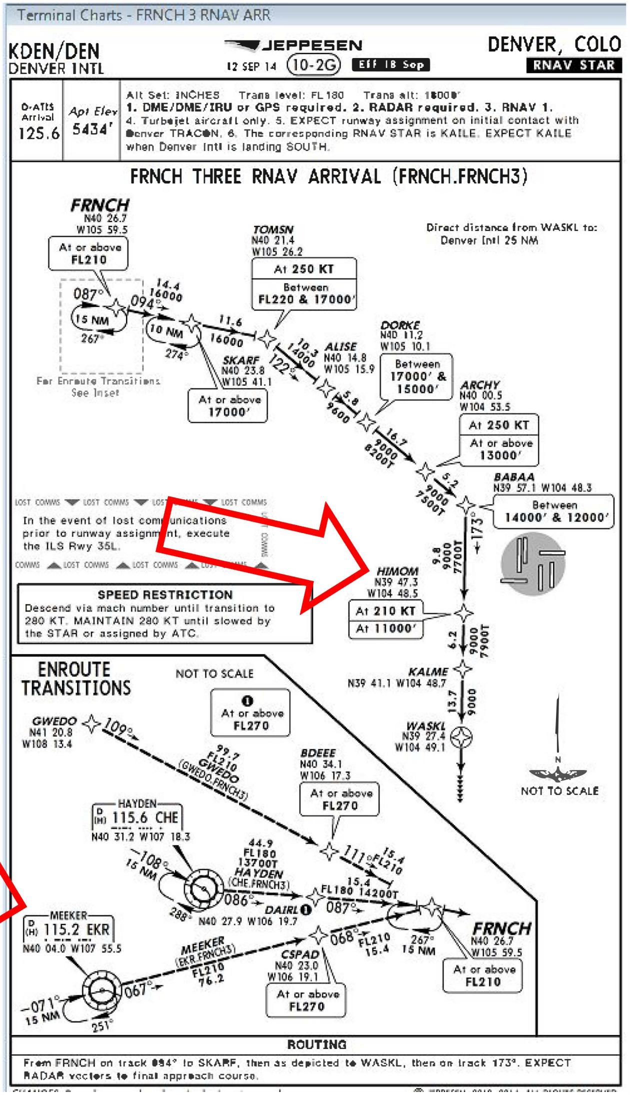
2015 – KDEN FRENCE THREE RNAV ARRIVAL



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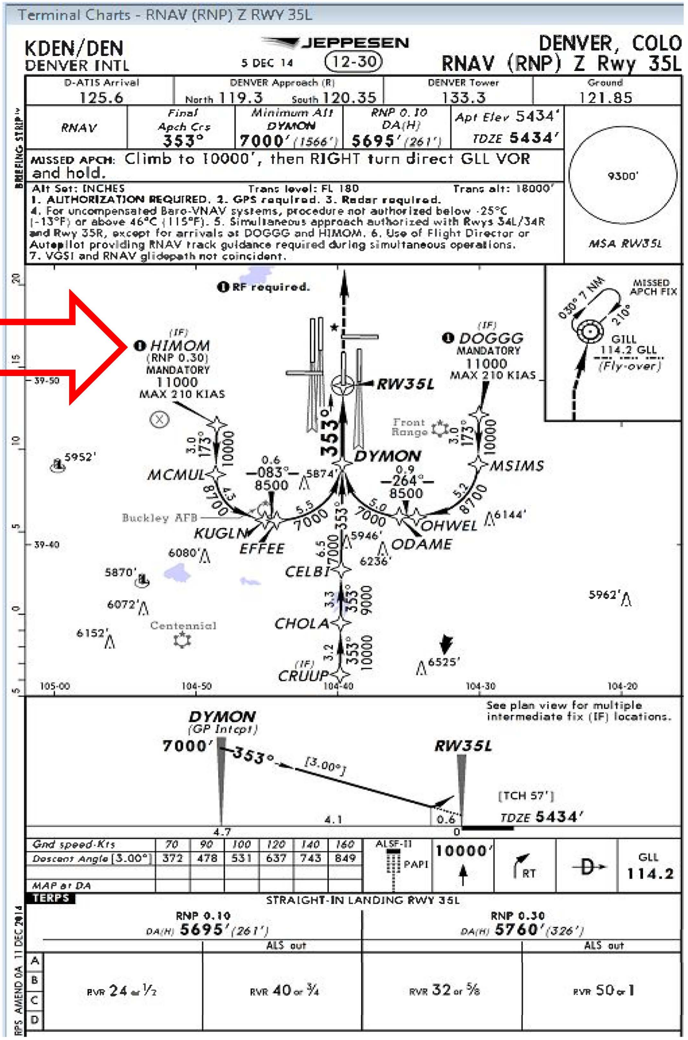
2015 – KDEN RNAV (RNP) RWY 35L



ARINC SPECIFICATION 429/702
Please record in .txt, csv, or Excel format (1 sec data).
A nominal update rate of more than one (1) time per second is unnecessary.
Error models should be OFF.

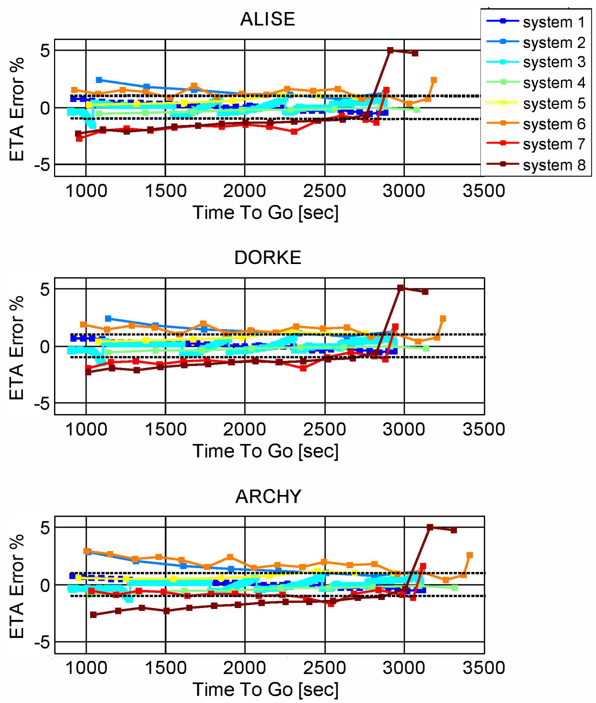
Data Element	Definition	Resolution
Time	GMT or Flight time	Seconds
Latitude	Simulation Aircraft State (truth)	Seconds
Longitude		Seconds
Barometric Altitude		Feet
Indicated Airspeed		Knots
Ground Speed		Knots
Track Angle		Degrees
Roll Angle		Degrees
Vertical Speed		Feet per minute

Flight Plan Fixes in order………

- MELEC → WOMEX → NSIGN → COKKE → MURFI → UPJA → BORZI → EKR → CSPAD (approximate top of descent point), → FRNCH → SKARF → TOMSN → ALISE → DORKE → ARCHY → BABAA → HIMOM → MCMUL → KUGLN → EFFEE → DYMON

Sample Results: Relative Error

- **The horizontal dashed lines represent the 1% of time-to-go requirement, providing a visual look at compliance of the various systems as the flight proceeds (right to left).**



ETA <u>Relative Error</u> for Fixes: ALISE, DORKE, ARCHY

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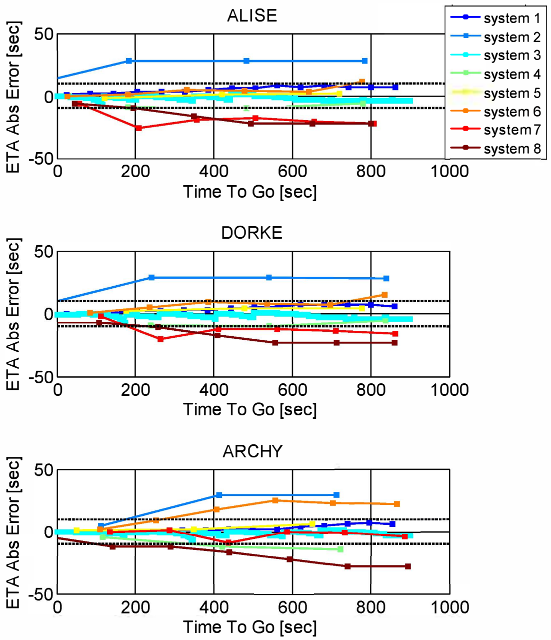
Sample Results: Absolute Error

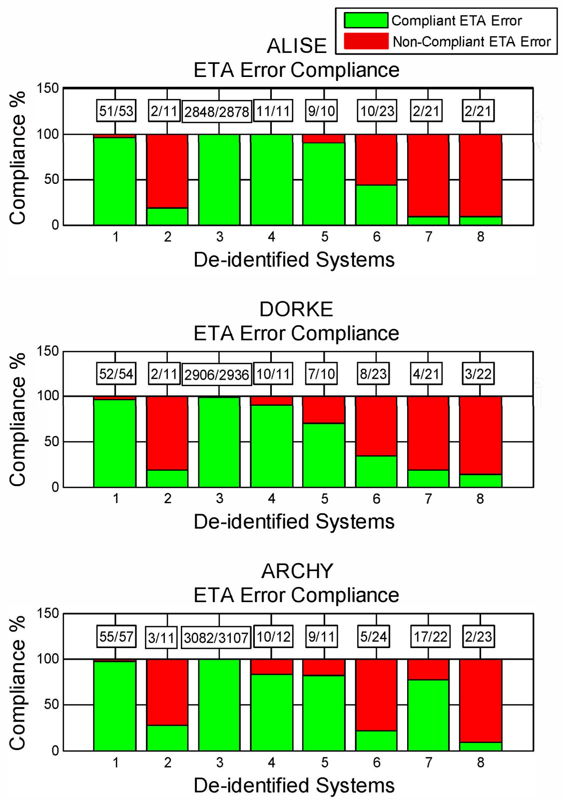
- **The horizontal dashed lines in these figure represent the 10 second requirement for time-to-go less than 15 minutes, providing a visual comparison of the ETA errors to the requirement as the flight proceeds (right to left).**



ETA <u>Absolute Error</u> for Fixes: ALISE, DORKE, ARCHY

Sample Results: ETA %Compliance

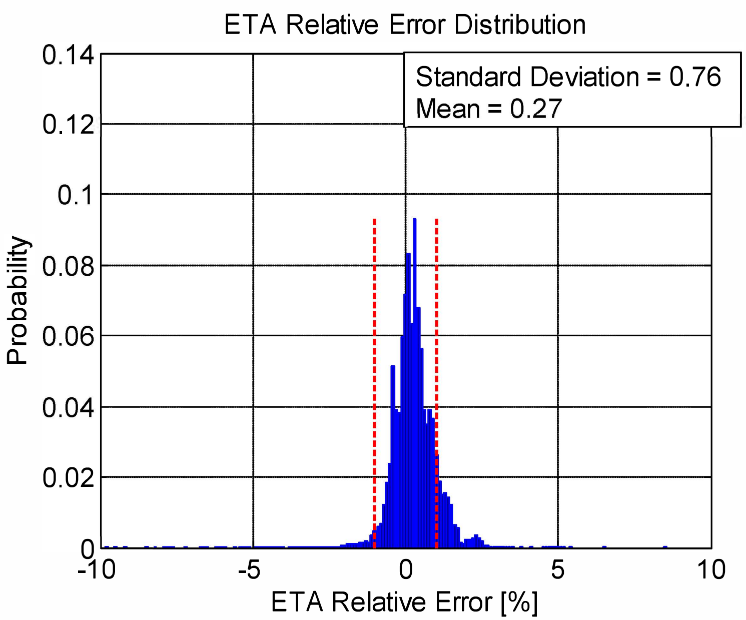
- **This sample shows ETA %Compliance at three fixes for each system. The horizontal axis represents the systems (1 – 8) and the vertical axis ETA %Compliance. The ratio over each bar shows the number of compliant ETAs over the Total Number of ETAs. The portion of the bar in green shows %compliant, while the red portion shows the %non-compliant.**



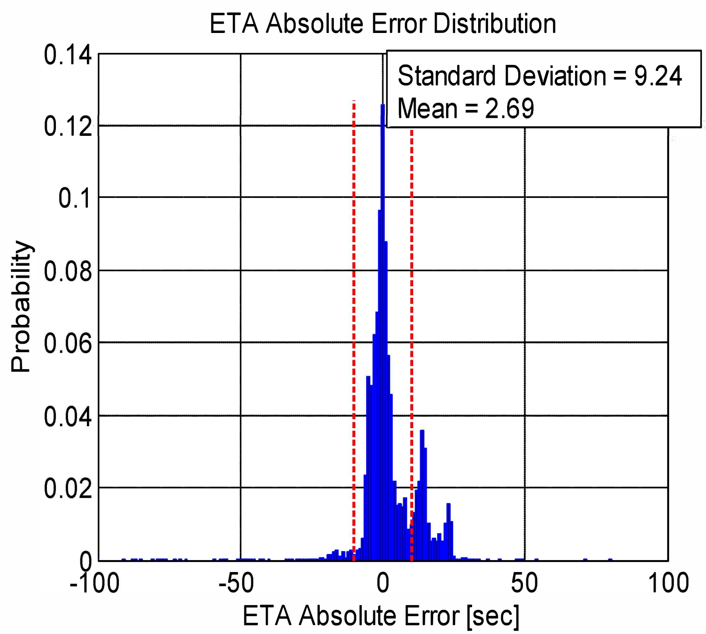
ETA Compliance for Fixes: ALISE, DORKE, ARCHY

Overall Compliance Distributions

- **The overall behavior of the ETA Relative and Absolute Error can be visualized in the histograms below. Both error distributions (relative and absolute) are centered close to zero, which is a desired characteristic. The standard deviation for each distribution (0.75% for relative error and 9.24 seconds for absolute error) is within the requirements (1% for relative error and 10 seconds for absolute error).**



Relative



Absolute

Conclusions

- **Considering all estimates at all fixes across all systems, the standard deviation of ETA error is within the performance requirement for both absolute and relative errors**
- **Considering individual systems across the full flight plan,**
  - there is little effect on ETA error either from the flight phase where the fix is located OR the flight phase where the aircraft was when the ETA was calculated
  - With times to go > 20 minutes, ETA errors are relatively consistent and less than 1.5%
  - Several systems meet the 10 sec for the full 10 minutes where it applies and most meet it for at least 7 minutes
- **Overall, it appears that lateral and vertical profiles are integrated well in all systems, it appears that wind models may account for most of the non-compliance**

Recommendations

- Based on the compliance demonstrated in these tests, we believe improvements may be needed to move further toward the FAA 2015 NextGen Implementation Plan.
- As future upgrades are made to FMSs, we would recommend that manufacturers revisit their:
  - Level of lateral / vertical path integration, as well as
  - Internal wind models with the goal of improving the performance of ETA predictions.
- We would also recommend that as part of the NextGen evolution:
  - Bounds be assessed for ETA performance using current systems, and
  - Mechanisms for utilization of airborne ETA information be considered to augment ground system predictions.

Photo by Captain Al Herndon, TWA

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